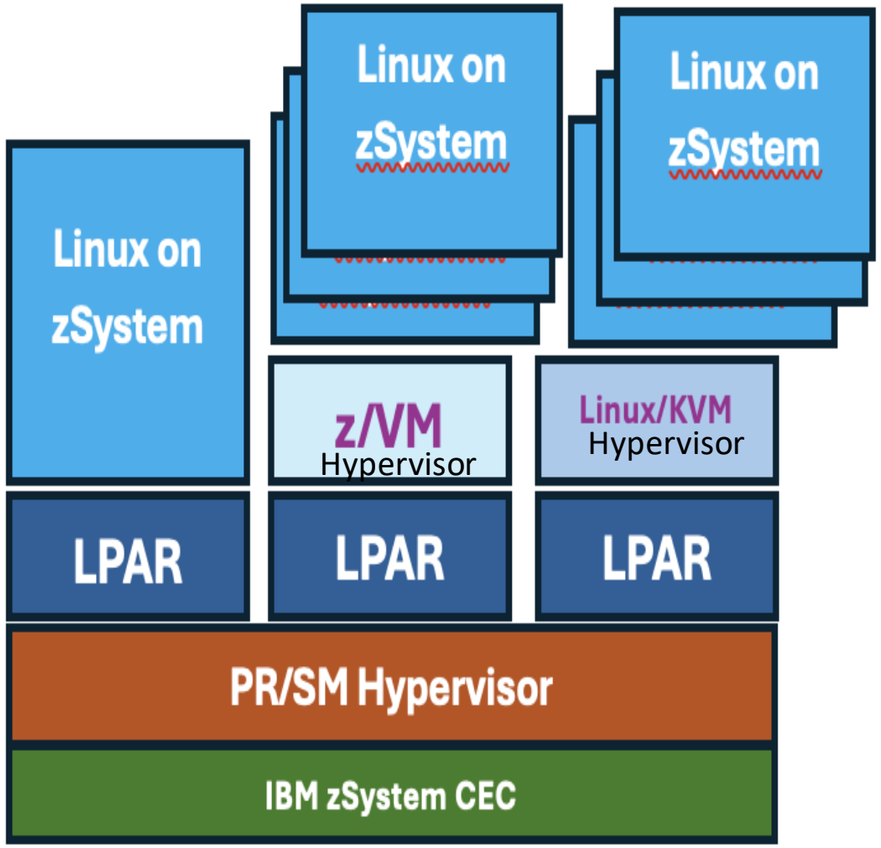
Demystifying Hypervisor Performance with a Case Study of Network performance

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Hypervisors

Virtualize the Hardware resources enabling multiple instances of similar or different operating systems to be run on a single set of computing resources.

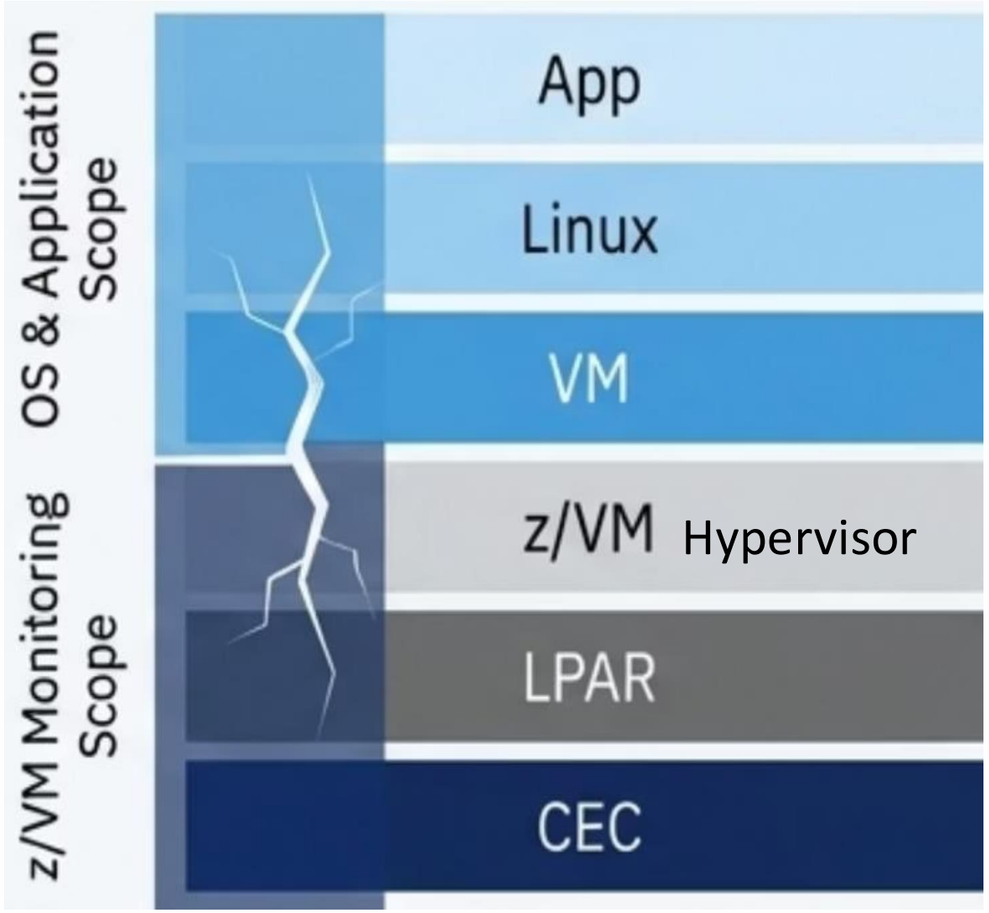


Benefits of Hypervisor technology

- Efficient use of large enterprise systems
- Enables server consolidation
- Isolation of resources
- Reduction in operational cost etc.

Performance Analysis of virtualized systems require deeper understanding of all the layers of the solution

Standard monitoring tools create data silos, limiting visibility across the stack



The challenge is that performance data is often scoped to a specific layer

- **OS & Application Scope:** Tools running within a Linux guest see virtual resources. They can report on application behavior and OS-level metrics but are blind to the underlying hypervisor activity
- **z/VM Monitoring Scope:** Hypervisor-level tools see the real hardware resource consumption but lack insight into the specific application processes running inside the guest VMs

Relying on one scope alone leads to incomplete analysis and incorrect conclusions

How to arrive at an unified view of system performance?



Co-relate performance monitoring data from hardware, hypervisor, operating system, application

Case Study: Identifying a bottleneck by Correlating Hypervisor & OS Data

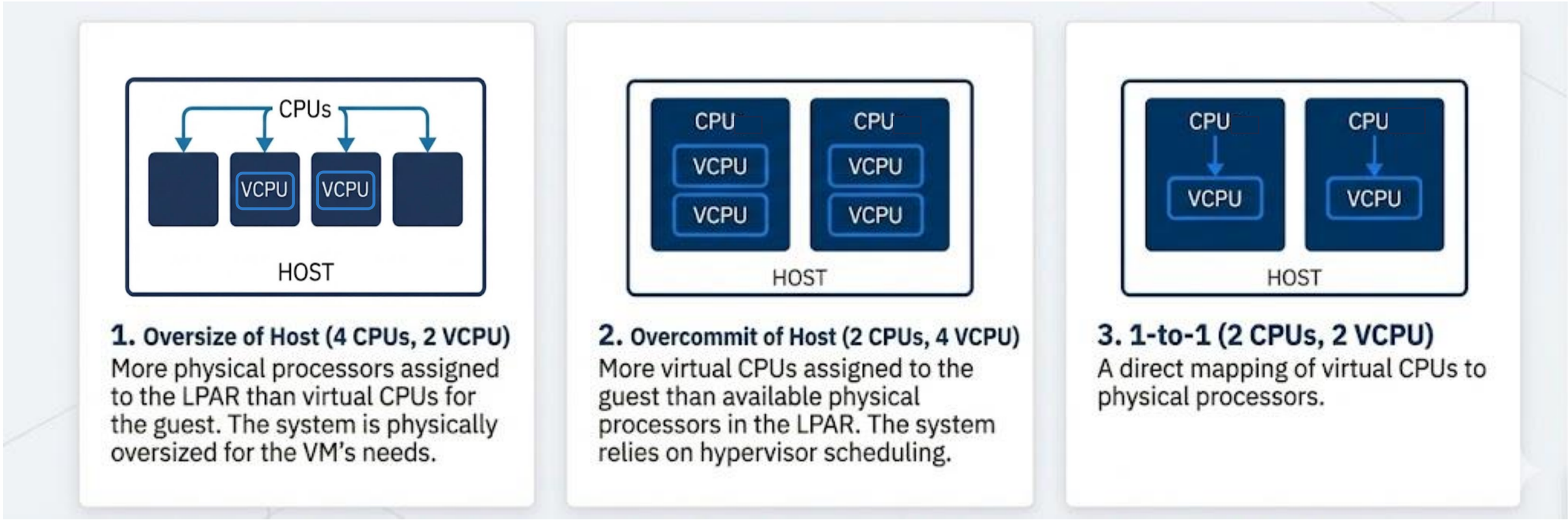
End-to-end approach to a common performance tuning scenario: **Optimizing CPU and Network configurations**

Experiment

Analyze the impact of CPU and virtual CPU (vCPU) configurations on network bandwidth

Goal

Determine the optimal configuration and understand the performance limitations of each setup

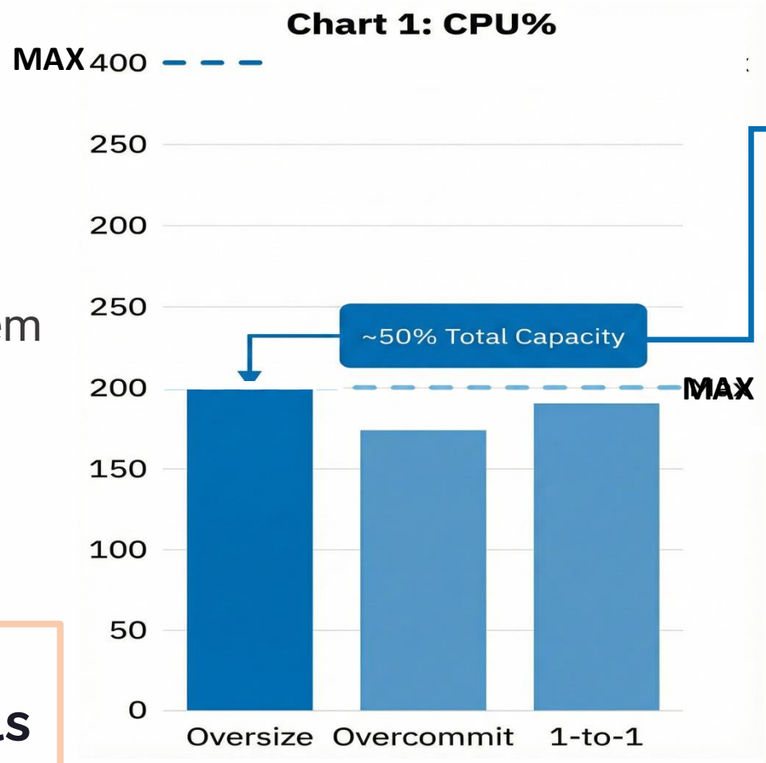


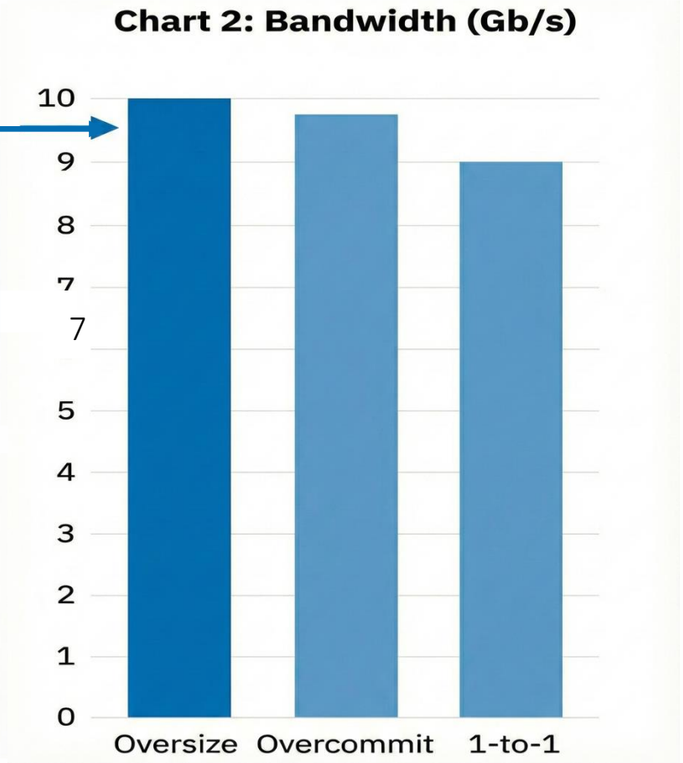
Looking only at hypervisor level metrics, the results are counterintuitive

- Oversize configuration (4 CPU, 2 VCPU) achieves the highest network bandwidth
- However, its CPU usage is capped at ~50%, System isn't using the two extra CPUs

The Puzzle: *Why can't a system with four available processors drive its two virtual CPUs past 50% utilization to achieve even higher bandwidth?*

Correlating both hypervisor data (mondata) and OS-level data(sar) reveals the true bottleneck: Network contention





- **Key Insight:** As network connections increase, **CPU Steal Time rises steadily**, indicating contention for resources. The vCPUs are waiting.

- **The Answer:** The network interface utilization peaks at **97%**. The system isn't bottlenecked by CPU at all; it's bottlenecked by the network bandwidth. The CPUs are idle because they are waiting on a saturated network channel.

SAR & MonData Analysis Summary

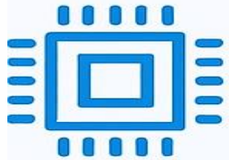
	CONFIG	Connections	Steal Time%	Interface Util%
Oversize	4 CPU, 2 VCPU	24	8.8	37
	4 CPU, 2 VCPU	56	20.99	82
	4 CPU, 2 VCPU	**108**	**23.67**	**97**
Overcommit	2 CPU, 4 VCPU	108	25	85
1-to-1	2 CPU, 2 VCPU	108	26	82

The End-to-End analysis provides a definitive diagnosis for each configuration.

The holistic view allowed us to correctly identify the limiting factor for each scenario:



4 CPU, 2 VCPU (Oversize) was bottlenecked by network bandwidth. The CPU had headroom it couldn't use.



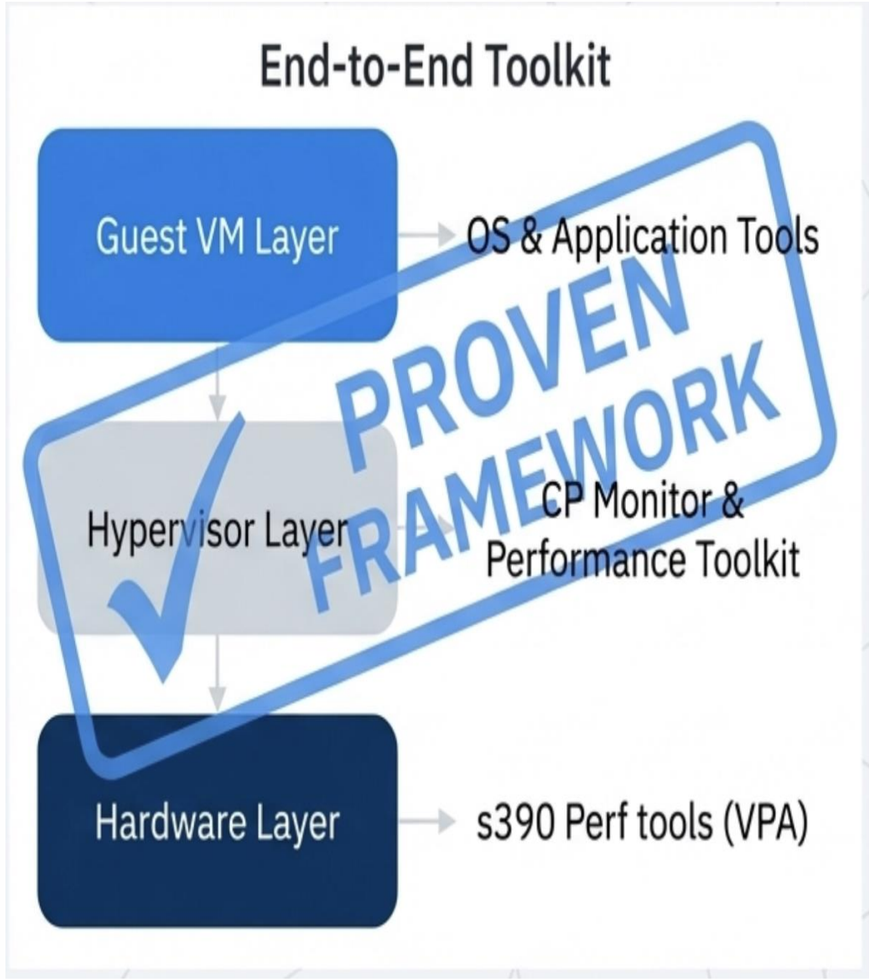
2 CPU, 4 VCPU (Overcommit) and **2 CPU, 2 VCPU (1:1)** were bottlenecked by **CPU**, preventing them from fully saturating the network channel.

Conclusion: Without combining data from both layers, we would have incorrectly diagnosed the Oversize configuration as a CPU problem.

The End-to-End Performance View is the key to accurate analysis

Effective performance engineering on z/VM is a Systematic process of looking across all layers of the technology stack

- **Start with a holistic view**
- **Use the right tool for each layer to gather data**
  - **to uncover the true relationships between components and identify the root cause of performance issues**



In a complex, virtualized environment, performance data from a single layer is just one chapter. **A multi-layered, end-to-end view transforms performance analysis from guesswork into a data-driven science, enabling true optimization and mastery of the zSystem platform.**

Hardware-sampling based performance measurements (like s390x Perf Tools) combined with visual analyzers offer hierarchical views and reports that pinpoint where runtime is spent, enabling precise hardware-level performance diagnostics.

Thank you !

Q & A